

Find the Percent of a Number

Lesson 4-4

Name: _____

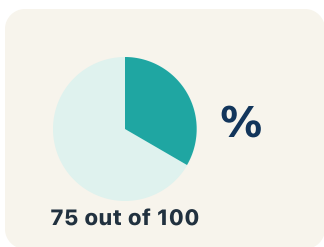
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

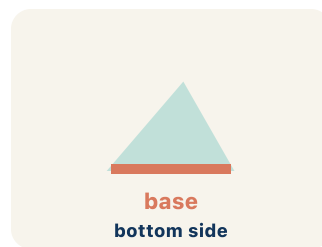
Picture first, then the word, then a plain-language meaning. Say each word out loud.



30% means 30 out of 100

Percent

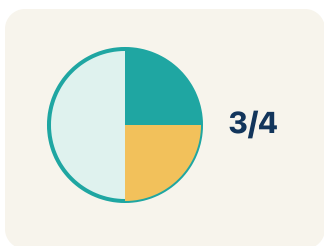
Write the definition:



In '25% of 80,' the base is 80

Base

Write the definition:



25% of 80 = 20; the part is 20

Part

Write the definition:

$$x + 2 = 7$$

balanced with =

part = percent $\times$ base

Equation

Write the definition:

~~\$40~~

\$28

30% off

\$40 shirt at 25% off: discount = \$10

Discount

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the percent of a number using an equation or a model.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Percent

Base

Part

Equation

Discount

1

A number out of 100 is a _____.

2

In a percent problem, the whole amount you take the percent of is the _____.

3

The piece of the whole that the percent represents is the _____.

4

A math sentence with an equal sign, like $\text{part} = \text{percent} \times \text{base}$, is an _____.

5

An amount subtracted from a price is a _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is 20% of 150?

- A. 30
- B. 20
- C. 35
- D. 75

Step 1 $20\% = 0.20$.

Step 2 Multiply: $0.20 \times 150 = 30$.

 **Answer:** A. 30

 We do — together

Solve this one with your class using the same steps.

Problem: What is 50% of 84?

- A. 42
- B. 50
- C. 34
- D. 168

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: What is 10% of 320?

- A. 32
- B. 3.2
- C. 320
- D. 64

Show your work:

| Try It

Solve on your own. Check the answer key when you are done.

1. Your bowling score was 60 points. The bonus round adds 15% more points. How many bonus points is that?

- A. 9
- B. 15
- C. 45
- D. 75

Show your work:

2. A snack at the arcade costs \$40. The sales tax is 5%. How many dollars is the tax?

- A. \$2
- B. \$5
- C. \$8
- D. \$20

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A class has 32 students. 25% are in the math club, 50% are in the reading club, and 12.5% are in both. How many students are in each group? Can the total of all three groups be more than 32? Explain why or why not.

Sentence starter: There are ____ in math club, ____ in reading club, and ____ in both. The total ____ be more than 32 because ____.

Show your work:

Reflect — Exit Ticket

What is 45% of 360?

- A. 162
- B. 145
- C. 180
- D. 126

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Percent
2. **Notes 2:** Base
3. **Notes 3:** Part
4. **Notes 4:** Equation
5. **Notes 5:** Discount
6. **We Try (notes):** A. $42 - 50\% = 0.50$. Multiply: $0.50 \times 84 = 42$. (Or simply divide by 2: $84 \div 2 = 42$.)
7. **You Try (notes):** A. $32 - 10\% = 0.10$. Multiply: $0.10 \times 320 = 32$. (Or move the decimal one place left.)
8. **Try It 1:** A. $9 - 15\% = 0.15$. Multiply: $0.15 \times 60 = 9$ bonus points.
9. **Try It 2:** A. $\$2 - 5\% = 0.05$. Multiply: $0.05 \times 40 = \$2$ in tax.
10. **Exit Ticket:** A. $162 - 45\% = 0.45$. Multiply: $0.45 \times 360 = 162$.